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FOCUS REVIEW

A computational method for characterizing molecular-scale load transfer in polymer systems with structural heterogeneity

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Abstract

Contemporary polymer physics emphasizes polymer material design, which uses nano- and microscale structures to predict and optimize material properties. Despite their importance, predicting the mechanical behavior of polymers remains challenging because of the diverse configurations of their molecular chains. In recent years, quantitative structure–property relationship modeling based on molecular dynamics (MD) simulations has become increasingly important. MD simulations excel at resolving sub-10-nm-scale morphological features, providing critical insights into network topology, chemical conformation, and molecular transitions. This review highlights recent MD simulation studies that have focused on subcontinuum heterogeneities in polymers from the perspective of their mechanical properties. The theoretical framework for rationally distributing the stress tensor to individual molecular components is revisited, and the key achievements made via this approach are summarized.

Introduction

The precise design of the physical properties of polymers by considering their nanoscale and microscale morphologies is a critical focus in contemporary polymer physics. This area of study is substantially important in a variety of fields ranging from mechanical engineering to biology and materials engineering. Understanding the intricate relationship between the microscopic architecture and the resulting material properties offers considerable advantages, particularly in predicting and optimizing the performance prior to synthesis [1]. However, the mechanical behavior of polymeric systems is inherently complex due to the diverse structural configurations of the molecular chains, which introduces significant uncertainty in their property prediction.

To address these challenges, quantitative structure–property relationship modeling has been increasingly pursued via computational methods based on molecular dynamics (MD) simulations [2–4]. Traditional MD simulations have been noted for their ability to precisely control design parameters and rapidly accumulate property data through massive parallel computations. As a result, early research efforts focused primarily on the rational modeling and description of molecular and supramolecular structures [5–7]. However, the scope of MD simulations has recently expanded to include the identification and quantification of physicochemical factors that determine macroscopic properties [8, 9]. This approach enables quantitative assessment of the individual contributions of molecular components during material characterization. Particularly noteworthy is the ability of MD simulations to resolve morphological heterogeneity at sub-10-nm scales [10–13], an achievement that remains challenging for experimental methods. These simulation techniques provide detailed insights into aspects such as network topology, chemical conformation, and molecular transition and are applied to effectively bridge continuum-scale physical properties, including strain energy, load transfer efficiency, and actuation performance [14–16].

In this focused review, we synthesize recent studies employing MD simulations to investigate subcontinuum-level heterogeneities in polymer systems from the

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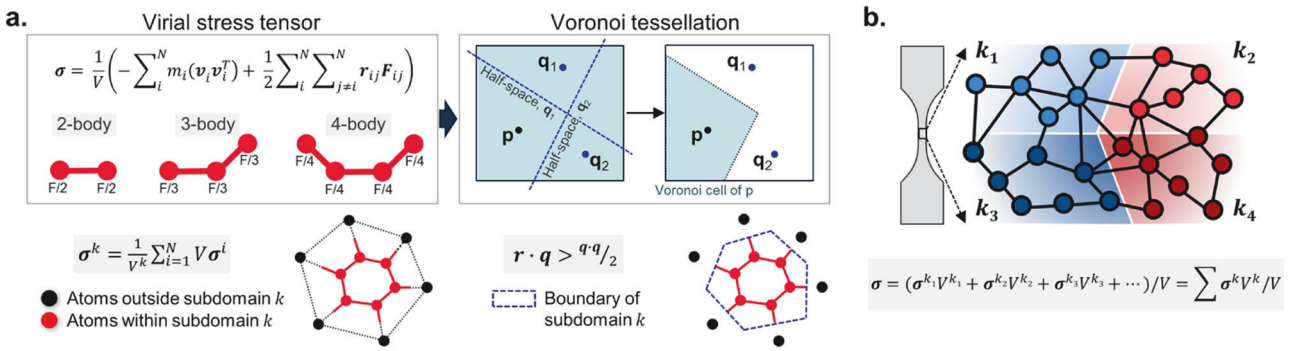


Fig. 1 Overview of the computational methodology. **a** Scheme of the virial stress tensor integrated with the Voronoi volume. **b** Determination of effective properties of macroscopic specimens from microscopic subdomain k

perspective of mechanical properties. We revisit the theoretical framework for rationally distributing the stress tensor of a system to its individual molecular components and summarize the key achievements of this approach. This focused review is structured as follows: After the methodology for evaluating molecular-level heterogeneity via molecular dynamics simulations is presented in Chapter 2, the application of the proposed methodology to various polymeric systems is performed in Chapter 3. "Mechanical behavior characterization of polymer-based nanocomposite interphases" summarizes the investigation of polymer nanocomposites functionalized with dissimilar fillers. The heterogeneous behavior of a spatially graded polymeric matrix due to interfacial interaction forces has been intensively explored. In "Understanding the mechanical behavior of crosslinked network microstructures" and "Molecular mechanism analysis of shape memory performance in epoxy networks", the methodology is extended to evaluate the mechanical behavior of network-structured polymers. The heterogeneous and asymmetric responses of molecular components classified by their crosslinking state or chemical structure are investigated during tensile and shape memory cycling. Finally, "Multiphysical behavior analysis of photoresponsive polymer nanocomposites" presents the multiphysical behavior analysis of photoresponsive polymer nanocomposites. While the matrix components undergo molecular kinking under external stimuli, the locally functionalized contractile behavior of the interfacial components due to interaction with the nanofiller is analyzed.

Methodology

The spatial characterization of atomistic systems begins with computation of the virial stress tensor (Fig. 1a, left). The virial stress, derived from the virial theorem, is governed by the following equation [17, 18]:

$$\boldsymbol{\sigma} = \frac{1}{V}\left(-\sum_{i}^{N_k} m_i(\boldsymbol{v}_i\boldsymbol{v}_i^T) + \frac{1}{2}\sum_{i}^{N_k}\sum_{j\neq i}^{N_k} \boldsymbol{r}_{ij}\boldsymbol{F}_{ij}\right), \tag{1}$$

where V is the volume of the unit cell; N_k is the number of constituent atoms; m_i and v_i are the mass and velocity of the i-th atom, respectively; and r_{ij} and F_{ij} indicate the relative position and interaction forces between the i-th and j-th atoms, respectively. This formulation determines the stress applied to atoms from the instantaneous velocities owing to thermal fluctuations and interatomic forces arising from covalent and noncovalent interactions. Unlike the macroscopic stress, which is derived from bulk measurements, the virial stress tensor explains the behavior and stress state of materials by considering atomic-scale interactions. Notably, Subramaniyan and Sun demonstrated that a sufficiently time-averaged virial stress tensor can be arithmetically equivalent to the mechanical Cauchy stress [19]. Therefore, virial stress allows for delicate extrapolation of continuum-scale properties from atomistic systems.

To obtain the local virial stress, the total virial stress within an atomistic system can be decomposed into contributions from individual atoms. This decomposition leverages the fact that the virial stress is inherently a per-atom quantity, allowing for the computation of an effective stress tensor for subdomain k via the following equation [20]:

$$\boldsymbol{\sigma}^k = \frac{1}{V^k}\sum_{i=1}^{N} V\boldsymbol{\sigma}^i, \tag{2}$$

where N is the number of atoms contained in subdomain k, V^k is the corresponding volume, and $\boldsymbol{\sigma}^i$ is the stress tensor of the i-th atom belonging to subdomain k. In other words, the effective stress for subdomain κ is obtained by simply summing the contributions of individual atoms and

normalizing them by their corresponding volumes. A critical aspect here is the rational determination of the effective volume of the subdomain. To do this, it is first necessary to understand the process by which the interatomic forces between pairs are distributed among the interacting atoms.

The force fields used for simulating polymer systems, including the CVFF, PCFF, and COMPASS, often require the consideration of up to four atoms to compute single interatomic forces (e.g., dihedral or improper quadruplets). Therefore, it is important to understand how the forces determined by multiple atoms are distributed to each entity. The potential part ($\boldsymbol{\sigma}_P$) of the virial stress presented in Eq. (1) can be rewritten as [21, 22]

$$\begin{aligned}\boldsymbol{\sigma}_P = &\frac{1}{2}\sum_{n=1}^{N_b}(r_n^1F_n^1 + r_n^2F_n^2)\\ &+\frac{1}{3}\sum_{n=1}^{N_a}(r_n^1F_n^1 + r_n^2F_n^2 + r_n^3F_n^3)\\ &+\frac{1}{4}\sum_{n=1}^{N_d}(r_n^1F_n^1 + r_n^2F_n^2 + r_n^3F_n^3 + r_n^4F_n^4)\\ &+\frac{1}{4}\sum_{n=1}^{N_i}(r_n^1F_n^1 + r_n^2F_n^2 + r_n^3F_n^3 + r_n^4F_n^4)\\ &+\frac{1}{2}\sum_{n=1}^{N_p}(r_n^1F_n^1 + r_n^2F_n^2) + \frac{1}{2}\sum_{n=1}^{N_c}(r_n^1F_n^1 + r_n^2F_n^2),\end{aligned} \tag{3}$$

where r and F are the relative positions and atomic forces of the pairwise interactions, respectively. where N_b, N_a, N_d, N_i, N_p, and N_c are the numbers of pairs in the bond, angle, dihedral, improper, van der Waals, and Coulomb interactions, respectively. As shown in Eq. (3), the virial stress indicates that the interatomic forces are evenly distributed among the participating atoms irrespective of the atomic species. In other words, determining the effective stress of subdomain k can be understood as equally splitting the contribution of interatomic interactions with its neighbors [23].

From this perspective, the effective volume of the subdomain can be appropriately determined via the Voronoi volume derived from the Voronoi tessellation (Fig. 1a, right) [24, 25]. The boundary of a Voronoi cell around a particular seed $\boldsymbol{p}$ is defined by a perpendicularly bisecting plane with its neighbors $\boldsymbol{q_1}$ and $\boldsymbol{q_2}$. It shares two fundamental characteristics with viral stress: (1) the Voronoi volume calculation does not apply weighting based on the atomic species; instead, the effective volume of an atom is determined solely by its relative position with respect to its neighbors. (2) Tessellation ensures that there are no missing or overlapping regions within the three-dimensional structure, meaning that the sum of the Voronoi volumes for all atoms is mathematically equivalent to the total volume of the unit cell. These properties align with the principles of local virial stress measurements, in which the effective stress is derived by partitioning the contributions from neighboring atoms.

Consequently, this methodology facilitates the precise determination of the effective properties of various subdomains within an atomistic system. The overall stress of the system was equivalent to the volume-fraction-weighted average of the stresses of the individual subdomains (Fig. 1b). This approach provides a robust framework for quantitatively assessing the contributions of distinct molecules or spatial regions under external loads. In the following sections, we review studies that have applied this methodology to explore atomistic systems in greater detail.

Application

Mechanical behavior characterization of polymer-based nanocomposite interphases

The most promising subject of this approach is the mechanical analysis of polymer-based nanocomposites [26–29]. Noncovalent interactions provided by heterogeneous fillers spatially grade the surrounding polymer matrix, significantly altering the macroscopic mechanical properties of the nanocomposites. Four factors are involved in determining the mechanical behavior of the crosslinked thermoset polymer interphase: (1) The dense polymer layer formed in the vicinity of the filler surface exhibits retarded chain dynamics, greatly suppressing energy dissipation during mechanical loading (Fig. 2a); (2) the absorbed polymer layer has a high local crosslinking density, which increases the load transfer efficiency [30]; (3) the van der Waals vacancies between the filler and matrix hinder effective load transfer across the interface; and (4) the fillers act as spatial obstacles that impede the formation of a continuous crosslinked network [31].

Owing to the interplay between these factors, the incorporation of a nanofiller has a dual effect; this also poses significant challenges for the precise characterization of the nanocomposite interphase. In this context, the subcontinuum approach achieved in MD simulations could be a promising strategy for quantifying the spatial properties of nanocomposites. Specifically, the effective contribution of the polymer matrix under uniaxial loading can be continuously quantified depending on the relative distance from the nanofiller in terms of the strain energy density (Fig. 2b).

The strain energy density, which quantifies the mechanical work done by a local subdomain k during

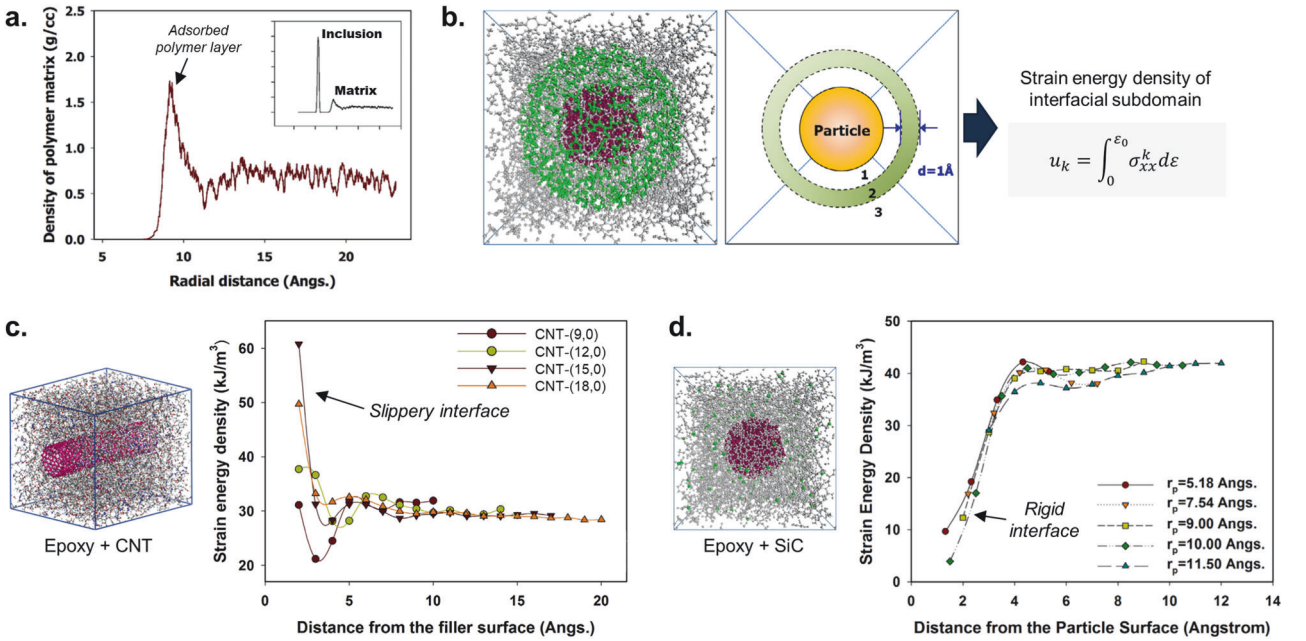


Fig. 2 Characterization of the interphase of polymer-based nanocomposites [26, 27]. **a** Typical radial density distribution of the polymer matrix interacting with a heterogeneous filler. **b** Calculation of the strain energy density distribution of the interfacial subdomain. **c** Strain energy density distributions of the epoxy matrix interacting with the CNTs and (**d**) SiC nanofiller. The CNT-(N,M) in (**c**) represents the chirality of the nanofiller. The r_p in (**d**) indicates the radius of the nanoparticle used. All figures are reprinted with permission from the publisher of the relevant literature

tension, is obtained from the following equation:

$$u_k = \int_0^{\varepsilon_0} \sigma_{xx}^k d\varepsilon \tag{4}$$

where σ_{xx}^k represents the virial stress component of subdomain k along the loading direction, ε represents the mechanical strain, and ε_0 represents the total uniaxial strain applied to the unit cell. To obtain the local load transfer characteristics at the nanoparticle/matrix interface, arbitrary local groups that have a spherical shell shape with a thickness of 1 Å and are concentric with the inserted nanoparticle are considered, as described in Fig. 2b. Then, the effective strain energy density for local regions ('1' is the part inside the area of interest, '2' is the area of interest with a thickness of 1 Å, and '3' is the part outside the area of interest) in the unit cell can be computed by partly summing the atomic virial stress of individual atoms in the unit cell and integrating them with the applied strain. This parameter quantifies the mechanical contribution of the local region during the tensile test. Therefore, the mechanical state of the interface due to interaction with nanoparticles, i.e., slipperiness or robustness, can be determined via this parameter.

Studies on highly crosslinked epoxies have revealed different responses depending on the type of nanofiller used (Fig. 2c, d). An epoxy matrix interacting with single-walled carbon nanotubes (CNTs) exhibited a relatively high strain energy density at the interface, which gradually decreased with increasing distance. This behavior was attributed to the insufficient adhesion provided by the slippery carbon hexagonal surface, which allowed the interfacial polymer chains to easily dislodge. These findings are consistent with those of other experimental studies that highlighted the low load transfer efficiency between carbon nanomaterials and polymers [32]. In contrast, the interaction of the epoxy with the SiC nanoparticles resulted in a reduced strain energy at the interface. Unlike CNTs, SiC nanoparticles form a rough surface due to deviations in atomic position, increasing friction with the surrounding matrix. Additionally, substantial partial charges of SiC nanoparticles foster robust adhesion with surrounding molecular components via polar interactions. Such suppressed chain dynamics and molecular redistribution owing to the formation of a robust interphase potentially contributed to the improved mechanical properties of the nanocomposites.

Overall, the mechanical heterogeneity between the interphase and bulk regions could be rigorously analyzed via a single simulation model. Importantly, in MD simulations, the properties of the interface and interphase are defined by interatomic interactions rather than by the user. This approach effectively circumvents the artifacts associated with contact condition specifications (e.g., the finite element method). Moreover, it facilitates extensive investigations of the interphase behavior under various filler parameters, including defects, oxidation, grafting, and functionalization [33].

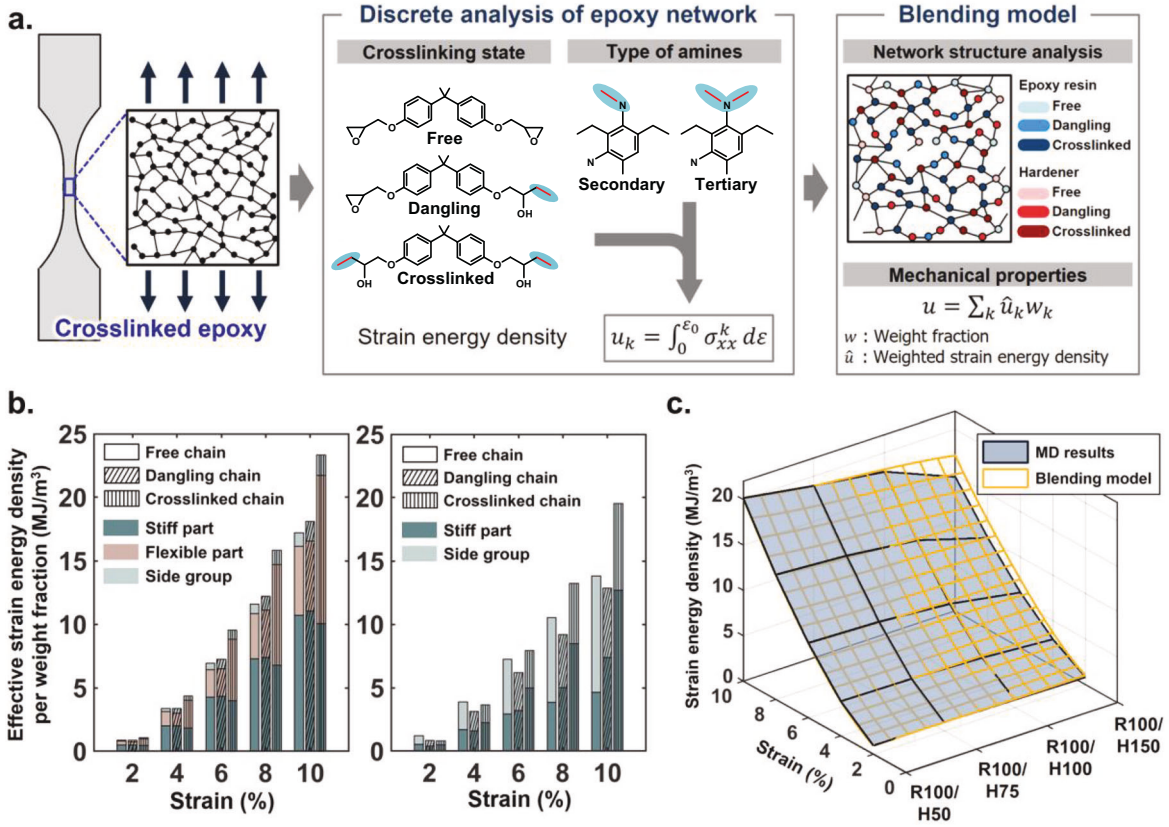


Fig. 3 Mechanical property prediction is based on thermoset network topology [14]. **a** Scheme of the topological modeling of crosslinked epoxy networks. **b** Strain energy density of the epoxy resin (right) and hardener (left) according to the crosslinking state. **c** Comparison of rule-of-mixture-based predictions and MD simulation results. All figures are reprinted with permission from the publisher of the relevant literature

Understanding the mechanical behavior of crosslinked network microstructures

The methodology was extended to identify topology–property relationships in crosslinked polymers [14]. Epoxy thermosets are among the polymers with the most active engineering applications owing to their excellent thermomechanical performance and ease of processing. The epoxide groups in the resin and the amines in the hardener undergo spontaneous crosslinking reactions during the curing process, forming a complex three-dimensional network. Here, the topology of the epoxy network was determined based on the bonding states of the molecular components (Fig. 3a). Specifically, the bonding state of a component can be classified as free, dangling, or crosslinked, depending on the number of crosslinks, which are homogeneously distributed to form an entangled network [34, 35].

The key consideration in this study is that even though the molecules have the same chemical structure, their inherent load-transfer efficiency varies depending on their crosslinking state. In other words, the individual molecules exhibited significant mechanical heterogeneity, and the macroscopic behavior of the entire sample was determined by their relative concentrations. Based on this framework, crosslinked epoxy microstructures were implemented in an all-atomistic environment. The mechanical contribution of the molecules was quantified in terms of the strain energy via virial stress, incorporating Voronoi tessellation during tension (Fig. 3b). These results suggest that epoxy molecules can effectively withstand loads only when both ends of the molecule are crosslinked. The possible molecular mechanisms include the following: (1) Crosslinked molecules facilitate efficient intramolecular load transfer via covalent bonds during macroscopic deformation of the network. (2) The restricted molecular mobility induced by crosslinking sites enhances the stability of intermolecular interactions, such as hydrogen bonding and π–π stacking of phenyl groups. These stabilized noncovalent interactions significantly influence the microscopic alignment of molecular bonds under tensile stress, thus enhancing resistance to external deformation [36]. In contrast, free and dangling segments dissipate the applied energy through intermolecular friction and therefore have lower mechanical resistance to loading. The effective responses of the hardener units were further characterized via a combination of amine types.

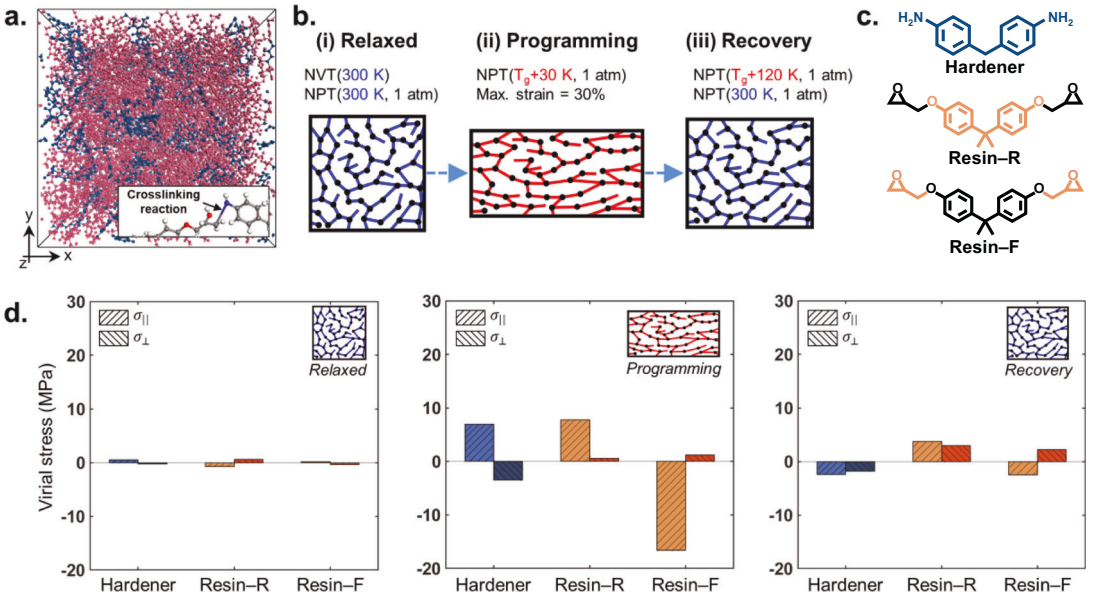


Fig. 4 Evaluation of the shape memory properties of thermoset epoxy networks [15]. **a** Molecular configuration of the cross-linked microstructure. Red and blue beads represent atoms that make up the epoxy resin and hardener, respectively. **b** Shape memory cycle reproduced in the MD environment. **c** Segmentation of epoxy resins and hardeners according to their chemical structures. **d** Virial stress accumulated for the molecular components at each stage. $\sigma_{||}$ and $\sigma_{\perp}$ represent the virial stress values along the stretching and normal directions, respectively. All figures are reprinted with permission from the publisher of the relevant literature

The relationship between the network topology and mechanical properties was analytically described via a rule-of-mixture-based blending model (Fig. 3c). The blending model provides a quantitative prediction of the strain energy of the entire network based on the strain energies of the individual molecules and their weight fractions. Assuming that all the molecules constituting the network have similar densities and are uniformly distributed, this approach was demonstrated to be effective for mechanical predictions based on the chemical composition of epoxy thermosets. In particular, as long as a single macromolecular network was formed during curing, the model provided accurate estimates of strain energy within 5% error. In other words, by considering the mechanical heterogeneity of the same molecule in different crosslinking states, a complete mechanical interpretation of the epoxy network can be achieved.

Molecular mechanism analysis of shape memory performance in epoxy networks

Generally, the shape memory performance of polymeric materials, including their actuation force and recovery time, is closely related to their intrinsic stiffness. In this regard, thermoset epoxy networks, which enable efficient load transfer through highly crosslinked structures, are promising candidates for shape memory materials with excellent performance. The external energy applied during the programming process was converted into the enthalpy of the molecular component through microscopic distortion of the covalent backbone [37, 38]. This stored enthalpy is released upon thermal activation above the glass transition temperature, promoting recovery of the material to its original shape. Therefore, understanding the underlying mechanism by which bond energy is stored during processing is important for the rigorous molecular-level design of shape memory polymer microstructures.

The shape-memory cycle of the epoxy thermoset was reproduced in the MD simulation environment (Fig. 4a, b) [15]. The resin and hardener molecules were inserted into a periodic unit cell to form a representative microstructure. The equilibrated model was heated above the glass transition temperature, followed by uniaxial stretching to a maximum strain of 30%. The model was subsequently cooled to room temperature while maintaining its deformed state. The programmed model was then exposed to high temperatures again, resulting in a reduction in axial length and recovery.

To elucidate the role of the chemical structure in energy storage, the epoxy components were categorized into three types (Fig. 4c): rigid resin containing a benzene ring (Resin-R), flexible resin composed of linear carbon chains (Resin-F), and a hardener component. The virial stress for each category was monitored throughout the shape-memory cycle. Notably, an asymmetric stress distribution within the network was observed depending on the chemical structure (Fig. 4d). Initially, all the components constitute a stress-free state. However, during uniaxial deformation, significant tensile stresses accumulate in

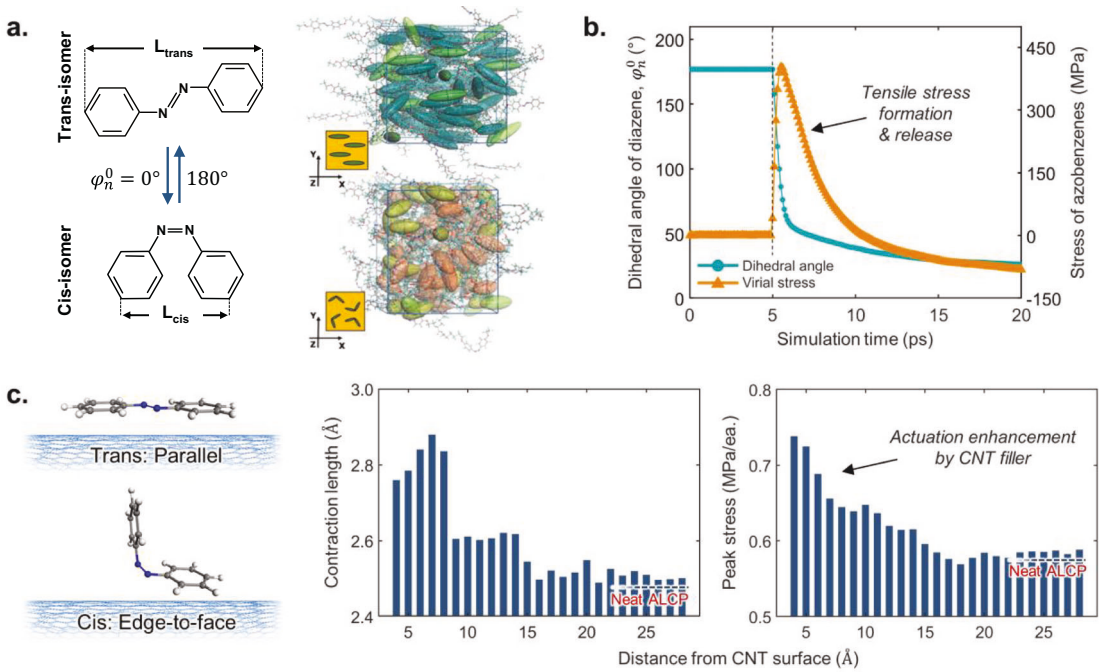


Fig. 5 Characterization of the photoinduced actuation behavior of ALCN [16, 45]. **a** Isomerization of azobenzene reproduced in an MD simulation cell. In the model, the ellipse represents the azobenzene moiety. **b** Stress tensor evolution in azobenzene during dynamic structural transition. The vertical dashed line indicates the point where isomerization occurred. **c** Changes in contraction length and maximum virial stress with distance from the CNT. All figures are reprinted with permission from the publisher of the relevant literature

the rigid components containing the benzene ring (i.e., Resin-R and Hardener). Upon removal of the applied load for shape fixation, a slight springback imparts residual compressive stress to the highly flexible hydrocarbon chains (i.e., Resin-F). This behavior suggests that the dihedral angles of the benzene rings, which possess a low energy barrier, are the primary sites for strain energy storage. The flexible components act as energy absorbers, buffering the system to maintain overall macroscopic equilibrium, despite the asymmetric stress distribution. In other words, strategies aimed at increasing the ring structure content within the network or reducing the binding energy barriers through chemical modifications can effectively improve shape memory performance. These asymmetric residual stresses are fully released as the molecular network undergoes significant rearrangement at high temperatures, generating the actuation force necessary for shape restoration.

Multiphysical behavior analysis of photoresponsive polymer nanocomposites

Azobenzene-functionalized liquid-crystalline networks (ALCNs) are a class of stimuli-responsive materials that exhibit macroscopic bending deformation upon exposure to ultraviolet (UV) light [39, 40]. The photoresponsivity is attributed to the conformational transitions of the photoactive azobenzene molecules embedded within the polymer matrix (Fig. 5a, left). Recent experimental studies have demonstrated that the incorporation of CNTs into ALCNs significantly enhances their photoactuation performance [41, 42]. However, the exact molecular mechanisms by which CNTs influence the structural dynamics of ALCNs during photoisomerization remain poorly understood, presenting challenges in elucidating their contribution to the observed mechanical strain. To address this, we conducted a detailed investigation of the structural behavior of azobenzene molecules in the vicinity of CNTs during trans-to-cis isomerization.

Upon UV irradiation, azobenzene molecules undergo isomerization, transitioning from a thermodynamically stable trans configuration to a metastable cis configuration. This reaction causes a significant reduction in the end-to-end distance of the azobenzene molecules—from 10 Å in the trans state to approximately 5.9 Å in the cis state—resulting in macroscopic contraction along the molecular axis [43]. Practical isomerization is inherently a quantum mechanical process involving complex transitions between electronic states. However, accurate modeling of the entire isomerization process, including excited-state dynamics and relaxation to the ground state, is computationally prohibitive for large-scale systems (>1000 atoms). To circumvent these challenges, Heinz et al. proposed a simplified approach that models the isomerization process by altering the equilibrium dihedral angle (φ_n^0) of the diazene group from 180° (trans) to 0° (cis) [44]. This approach allows for efficient simulation of the mechanical and operational

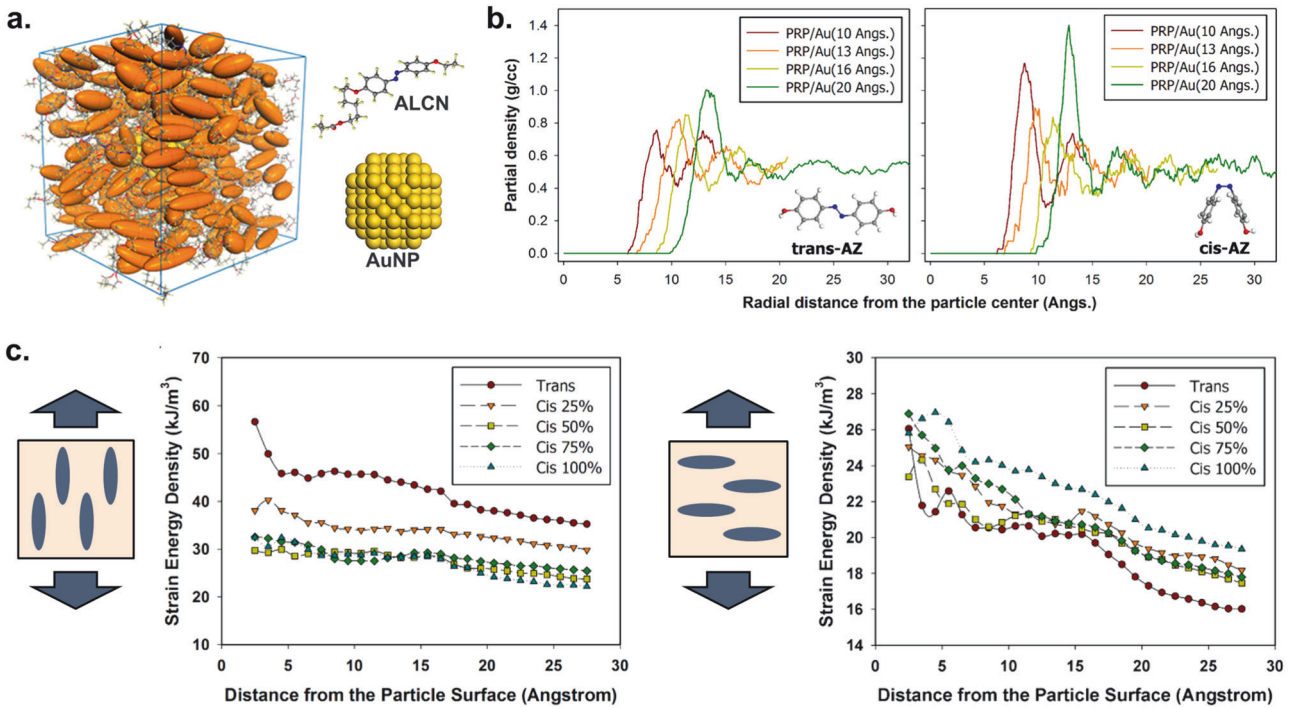


Fig. 6 Changes in the mechanical properties of ALCN nanocomposites are due to photoisomerization [48, 49]. **a** Molecular configuration of ALCN nanocomposites containing spherical nanoparticles. In the model, the ellipse represents the azobenzene moiety. **b** Radial density distributions of the polymer backbone constituents of the nanocomposites, including the trans isomer (left) and cis isomer (right). **c** Strain energy density distribution of the ALCP matrix according to the isomerization ratio. Mechanical loading was applied along the alignment direction (left) and perpendicular direction (right). All figures are reprinted with permission from the publisher of the relevant literature

properties of azobenzene-functionalized ALCNs within a nanometer-scale simulation cell (Fig. 5a, right) [45].

The key parameters of photoresponsive polymers include the actuation strain and photoinduced force. In this context, the photoinduced force of ALCNs can be quantified by evaluating the local virial stress tensor of azobenzene during isomerization (Fig. 5b) [16]. The rapid conformational transition from trans to cis induces substantial tensile stress within the azobenzene molecules; this is due to the structural resistance imposed by the crosslinked network. These optomechanical stresses decay exponentially as the diazene bonds reach a new equilibrium angle. Therefore, the spatial distribution of azobenzene actuation properties can be characterized by analyzing the variation in the peak stress.

The contraction length (L_{trans}-L_{cis}) and peak virial stress of the azobenzene moiety were evaluated as functions of distance from the CNT (Fig. 5c). The results indicated that both the contraction length and peak stress further increased upon interaction with the CNTs. In the trans state, azobenzene molecules are aligned parallel to each other, optimizing π–π stacking interactions with the CNT surface [46, 47]. However, upon isomerization, azobenzene adopts an edge-to-face configuration in which one benzene ring aligns perpendicular to the CNT axis. This transition not only amplifies the projected contraction length along the CNT axis but also significantly enhances the actuation stress. The effective range of the interaction effects of the CNTs (~15 Å) can also be identified from this analysis. Consequently, the methodology provides critical insights into the molecular origins of the enhanced macroscopic deformation in CNT-integrated ALCNs, facilitating a deeper understanding of their photomechanical behavior.

For the systematic design of photoresponsive polymer-based soft devices and actuators with mechanical rigor, a comprehensive understanding of their optomechanical behavior is essential. In this context, the tensile behavior of ALCN nanocomposites reinforced with spherical nanoparticles was studied as a function of the isomerization ratio (Fig. 6a) [48]. The interfacial response of the nanocomposites can be inferred from the spatial distribution of the backbone constituents (Fig. 6b). The incorporation of nanoparticles with strong physical absorption leads to the formation of a densely packed polymer layer near the nanoparticle surface. Notably, this densification was particularly pronounced when azobenzene isomerized to its cis state, as evidenced by the marked increase in the peak density of the absorbed polymer layer. In other words, unlike linear and rod-like trans-azobenzene, the curved configuration of cis-azobenzene facilitates optimal nanoparticle encapsulation.

The mechanical response of the polymer matrix under uniaxial loading was quantitatively evaluated via the strain energy density based on virial stress (Fig. 6c) [49]. The localized strain energy near the nanoparticle surface consistently exceeds that of the bulk material. This observation suggests that the thermodynamic incompatibility between the nanoparticles and ALCN results in facile detachment of interfacial molecules under mechanical stress. Importantly, as the isomerization ratio increased, the difference in energy density between the interface and the bulk decreased, irrespective of the loading direction. In other words, the transition of azobenzene to the cis isomer facilitates closer adsorption with the nanoparticles, thereby improving the mechanical stability of the nanocomposite interface.

Conclusion

Overall, the subcontinuum approach driven in an MD simulation environment is a powerful tool to elucidate the structure–property relationship by considering the microscopic heterogeneity of the polymer system. The integration of the virial stress with Voronoi tessellation provides a robust framework for distributing the overall stress tensor across individual atoms, making it an effective metric for the quantitative evaluation of the contributions of molecular components. This approach is particularly useful for identifying the key physicochemical factors that govern material properties. Current research efforts are extending this approach to explore more complex polymer physics, such as the distinct responses of crystalline and amorphous regions in semicrystalline systems or the contributions of individual phases in phase-separated polymers [50–53]. These advancements will provide crucial insights into the design of polymer materials at the microscopic and molecular levels.

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Compliance with ethical standards

Conflict of interest The authors declare no competing interests.

References

1. Le T, Epa VC, Burden FR, Winkler DA. Quantitative structure–property relationship modeling of diverse materials properties. Chem Rev. 2012;112:2889–919.
2. Theodorou DN. Understanding and predicting structure–property relations in polymeric materials through molecular simulations. Mol Phys. 2004;102:147–66.
3. Afzal MAF, Browning AR, Goldberg A, Halls MD, Gavartin JL, Morisato T, et al. High-throughput molecular dynamics simulations and validation of thermophysical properties of polymers for various applications. ACS Appl Polym Mater. 2020;3:620–30.
4. Nguyen D, Tao L, Li Y. Integration of machine learning and coarse-grained molecular simulations for polymer materials: physical understandings and molecular design. Front Chem. 2022;9:820417.
5. Okabe T, Oya Y, Tanabe K, Kikugawa G, Yoshioka K. Molecular dynamics simulation of crosslinked epoxy resins: curing and mechanical properties. Eur Polym J. 2016;80:78–88.
6. Bandyopadhyay A, Valavala PK, Clancy TC, Wise KE, Odegard GM. Molecular modeling of crosslinked epoxy polymers: the effect of crosslink density on thermomechanical properties. Polymer. 2011;52:2445–52.
7. Varshney V, Patnaik SS, Roy AK, Farmer BL. A molecular dynamics study of epoxy-based networks: cross-linking procedure and prediction of molecular and material properties. Macromolecules. 2008;41:6837–42.
8. Park H, Kim B, Choi J, Cho M. Influences of the molecular structures of curing agents on the inelastic-deformation mechanisms in highly-crosslinked epoxy polymers. Polymer. 2018;136:128–42.
9. Wick CD, Peters AJ, Li G. Quantifying the contributions of energy storage in a thermoset shape memory polymer with high stress recovery: a molecular dynamics study. Polymer. 2021;213:123319.
10. Li C, Strachan A. Evolution of network topology of bifunctional epoxy thermosets during cure and its relationship to thermomechanical properties: a molecular dynamics study. Polymer. 2015;75:151–60.
11. Zihan W, Peibin K, Tianyu W, Dongli C, Xiaoping Y, Gang S. Atomistic understanding of cross-linking network in different epoxy resin: effect of loop structure. Polymer. 2022;243:124629.
12. Radue MS, Jensen BD, Gowtham S, Klimek-McDonald DR, King JA, Odegard GM. Comparing the mechanical response of di-, tri-, and tetra-functional resin epoxies with reactive molecular dynamics. J Polym Sci Part B Polym Phys. 2018;56:255–64.
13. Bukowski C, Zhang T, Riggleman RA, Crosby AJ. Load-bearing entanglements in polymer glasses. Sci Adv. 2021;7:eabg9763.
14. Kim H, Choi J. Subcontinuum interpretation of mechanical behavior for cross-linked epoxy networks. Macromolecules. 2022;55:5916–25.
15. Kim Y, Kim H, Choi J. Mechanical origin of shape memory performance for crosslinked epoxy networks. Eur Polym J. 2023;194:112162.
16. Kim H, Choi J. Theoretical analysis of phase transition behavior of ALCP/CNT nanocomposites interface by photo and thermal stimulation. Compos A Appl Sci Manuf. 2023;175:107824.
17. Heyes DM. Pressure tensor of partial-charge and point-dipole lattices with bulk and surface geometries. Phys Rev B Condens Matter. 1994;49:755.
18. Thompson AP, Plimpton SJ, Mattson W. General formulation of pressure and stress tensor for arbitrary many-body interaction potentials under periodic boundary conditions. J Chem Phys. 2009;131:154107.
19. Subramaniyan AK, Sun C. Continuum interpretation of virial stress in molecular simulations. Int J Solids Struct. 2008;45:4340–6.
20. Reda H, Chazirakis A, Behbahani AF, Savva N, Harmandaris V. A methodology for determining the local mechanical properties of model atomistic glassy polymeric nanostructured materials. MethodsX. 2022;9:101931.
21. Reda H, Chazirakis A, Behbahani AF, Savva N, Harmandaris V. Mechanical properties of glassy polymer nanocomposites via atomistic and continuum models: the role of interphases. Comput Methods Appl Mech Eng. 2022;395:114905.

22. Barakat M, Reda H, Katsamba P, Shraim H, Harmandaris V. Predicting mechanical heterogeneity in glassy polymer nanocomposites via an inverse computational approach based on atomistic molecular simulations and homogenization methods. Mech Mater. 2024;197:105082.
23. Elder RM, Mattson WD, Sirk TW. Origins of error in the localized virial stress. Chem Phys Lett. 2019;731:136580.
24. Lu J, Lazar EA, Rycroft CH. An extension to Voro++ for multithreaded computation of Voronoi cells. Comput Phys Commun. 2023;291:108832.
25. Rycroft C. VORO++: a three-dimensional Voronoi cell library in C++. 2009.
26. Choi J, Shin H, Yang S, Cho M. The influence of nanoparticle size on the mechanical properties of polymer nanocomposites and the associated interphase region: a multiscale approach. Compos Struct. 2015;119:365–76.
27. Choi J, Shin H, Cho M. A multiscale mechanical model for the effective interphase of SWNT/epoxy nanocomposite. Polymer. 2016;89:159–71.
28. Reda H, Chazirakis A, Behbahani AF, Savva N, Harmandaris V. Revealing the role of chain conformations on the origin of the mechanical reinforcement in glassy polymer nanocomposites. Nano Lett. 2023;24:148–55.
29. Reda H, Chazirakis A, Savva N, Ganghoffer J-F, Harmandaris V. Gradient of mechanical properties in polymer nanocomposites: from atomistic scale to the strain gradient effective continuum. Int J Solids Struct. 2022;256:111977.
30. Sahraei AA, Mokarizadeh AH, George D, Rodrigue D, Baniassadi M, Foroutan M. Insights into interphase thickness characterization for graphene/epoxy nanocomposites: a molecular dynamics simulation. Phys Chem Chem Phys. 2019;21:19890–903.
31. Putz KW, Palmeri MJ, Cohn RB, Andrews R, Brinson LC. Effect of cross-link density on interphase creation in polymer nanocomposites. Macromolecules. 2008;41:6752–6.
32. Bortz DR, Heras EG, Martin-Gullon I. Impressive fatigue life and fracture toughness improvements in graphene oxide/epoxy composites. Macromolecules. 2012;45:238–45.
33. Kim H, Choi J. Interfacial and mechanical properties of liquid crystalline elastomer nanocomposites with grafted Au nanoparticles: a molecular dynamics study. Polymer. 2021;218:123525.
34. Li C, Strachan A. Free volume evolution in the process of epoxy curing and its effect on mechanical properties. Polymer. 2016;97:456–64.
35. Sridhar AS. Effect of stoichiometry on crosslinked epoxy resin characteristics: structural heterogeneities, topological defects, properties, free volume and segmental mobility. Soft Matter. 2022;18:2354–72.
36. Shudo A, Izumi A, Hagita K, Nakao T, Shibayama M. Structure-mechanical property relationships in crosslinked phenolic resin investigated by molecular dynamics simulation. Polymer. 2017;116:506–14.
37. Yan C, Li G. A mechanism-based four-chain constitutive model for enthalpy-driven thermoset shape memory polymers with finite deformation. J Appl Mech. 2020;87:061007.
38. Fan J, Li G. High enthalpy storage thermoset network with giant stress and energy output in rubbery state. Nat Commun. 2018;9:642.
39. Kim H, Ahn S-K, Mackie DM, Kwon J, Kim SH, Choi C, et al. Shape morphing smart 3D actuator materials for micro soft robot. Mater Today. 2020;41:243–69.
40. Pang X, Lv JA, Zhu C, Qin L, Yu Y. Photodeformable azobenzene-containing liquid crystal polymers and soft actuators. Adv Mater. 2019;31:1904224.
41. Kim H, Lee JA, Ambulo CP, Lee HB, Kim SH, Naik VV, et al. Intelligently actuating liquid crystal elastomer-carbon nanotube composites. Adv Funct Mater. 2019;29:1905063.
42. Bi M, He Y, Wang Y, Yang W, Qin B, Xu J, et al. Photo actuation performance of nanotube sheet incorporated azobenzene cross-linked liquid crystalline polymer nanocomposite. Polymers. 2019;11:735.
43. Rau H. Photoisomerization of azobenzenes. Photochem. Photophys. 1990;2:119–41.
44. Heinz H, Vaia R, Koerner H, Farmer B. Photoisomerization of azobenzene grafted to layered silicates: simulation and experimental challenges. Chem Mater. 2008;20:6444–56.
45. Choi J, Chung H, Yun J-H, Cho M. Photo-isomerization effect of the azobenzene chain on the opto-mechanical behavior of nematic polymer: A molecular dynamics study. Appl Phys Lett. 2014;105:221906.
46. Kim H, Kim H, Choi J. Interface mechanics of liquid crystal polymer nanocomposites with high concentrations of MWCNTs. Compos Sci Technol. 2022;222:109376.
47. Kim H, Choi J. Mechanical assessment of interfacial stability of LCP/MWCNT nanocomposites during phase transition. Compos A Appl Sci Manuf. 2023;167:107461.
48. Choi J, Chung H, Yun J-H, Cho M. Molecular dynamics study on the photothermal actuation of a glassy photoresponsive polymer reinforced with gold nanoparticles with size effect. ACS Appl Mater Interfaces. 2016;8:24008–24.
49. Choi J, Shin H, Cho M. Multiscale multiphysical analysis of photo-mechanical properties of interphase in light-responsive polymer nanocomposites. Compos Sci Technol. 2018;160:32–41.
50. Kim H, Choi J. A lamellar-morphology-based computational modeling for predicting the thermal conductivity of semicrystalline polymers. Int J Mech Sci. 2024;282:109622.
51. Kim H, Choi J. Influence of crystalline structure on creep resistance capability in semi-crystalline Polymers: a coarse-grained molecular dynamics study. Int J Fatigue. 2024;188:108517.
52. Kim H, Won D, Ko SH, Choi J. Thermodynamic mechanism governing the coalescence of conductive particles in PEDOT: PSS under laser irradiation. Macromolecules. 2024;57:2048–56.
53. Ma R, Wang Y, Huang H, Zhao X, Li X, Zhang L, et al. Insight into the fracture properties and molecular mechanism of polyurethane. Macromolecules. 2024;57:4484–94.

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